

Project Proposal



**A unified platform driving Sri Lanka's circular economy
through reselling, renting, and donating.**

**Prepared by CS - 31
Second – Year Group Project**

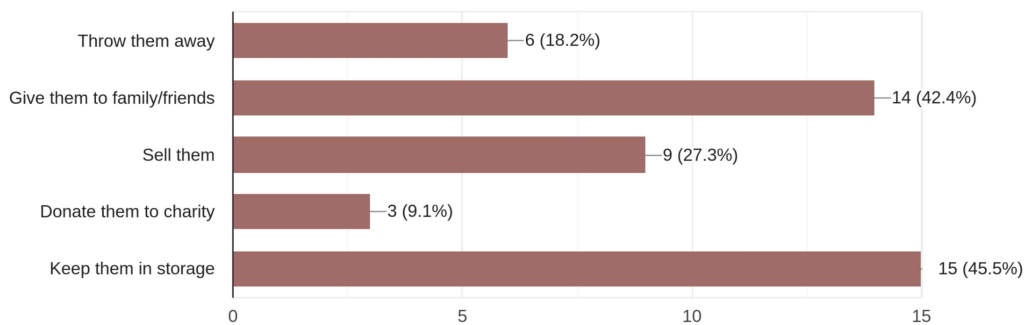
1. Introduction to the Project

This system is a web-based marketplace designed to manage underused or idle resources by allowing individuals to resell, rent, or donate items they no longer use frequently. It connects people who have usable items with those who need them, creating a shared platform that promotes efficient resource utilization.

Lack of a proper system to share, resell, rent, or donate underused items leads to many usable resources remaining idle and wasted. As a result, people who need such items are forced to buy new ones, increasing unnecessary spending and inefficient use of available resources. Currently, there is no proper system to circulate such underused items, leading to a significant amount of usable resources remaining idle or wasted. Studies suggest that households use only around 20–30% of the items they own regularly, while the rest remain unused. At the same time, many individuals purchase new products due to lack of access to these idle resources, resulting in unnecessary spending and inefficient resource distribution.

What do you currently do with items you no longer need?

33 responses



This system helps reduce waste by maximizing the use of existing resources, lowers costs for users by providing affordable access to items, and promotes a culture of sharing and sustainability. It also creates opportunities for individuals to earn value from unused items while supporting a more efficient and responsible consumption model.

Alternatives in Other Countries

Several countries already use platforms that support similar ecosystems:

- Facebook Marketplace – Allows users to buy and sell used items locally.
- eBay – A global platform mainly focused on reselling used and new goods.
- OLX – Popular in many countries for classified ads and second-hand items.
- Airbnb – Enables short-term renting, mainly for accommodation.
- Turo – A peer-to-peer vehicle rental platform.
- Freecycle – Focuses on donating items for free within communities.

Core Features Comparison

Feature Category	NexUse	ikman.lk	FB Marketplace	Daraz.lk	karuna.lk
Second Hand Selling	✓	✓	✓	✗	✗
Manageable Renting System	✓	Limited (houses and vehicles)	✗	✗	✗
Sharing (Free, Temporary)	✓	✗	✗	✗	✗
Multiple Exchange Types	✓	✗	✗	✗	✗
In-App chat system	✓	✗	✓	✓	✗
Donation System	✓	✗	✗	✗	✓

2. Project Goal

The primary goal of this project is to develop an integrated web platform that enables users to exchange underutilized items by selling, renting, sharing, or donating them. The system aims to improve resource utilization by making idle items accessible to others who need them.

Key objectives:

- Enable users to sell items they no longer need
- Allow users to rent out items for temporary use
- Provide an option to share or donate items freely
- Reduce the number of unused or wasted resources
- Offer affordable access to items for other users
- Support multiple exchange methods within a single platform instead of using separate systems.
- Provide a simple and user-friendly system for listing, searching, and requesting items.

3. Scope of the project

- **Actors**

- Admin
- Reseller
- Buyer
- Donar
- Donation Receiver
- Lender
- Renter

- **In-Scope**

- The system supports user registration, login, profile management, and reviews, and allows users to report disputes without providing resolution mechanisms.
- The system enables users to list any type of item with details such as description, condition, and availability type, while maintaining ownership rules based on the type of transaction.
- The system allows users to request items for buying, renting, sharing, or donation, with item owners having the ability to accept or reject requests without automatic approvals.
- The system tracks rental and sharing periods with start and return dates, sends reminders, manages late returns with penalties, verifies item condition, and allows compensation requests for damage or non-return.
- The system supports selling with ownership transfer, donating as a free permanent transfer, and sharing as a free temporary exchange with return tracking.
- The system allows users to list items for disposal, which are handled by authorized parties with mutually agreed terms, while the platform only supports coordination.
- The system provides notifications for requests, approvals or rejections, return reminders, and late return alerts.
- The system enables users to rate and review other users involved in transactions, with ratings visible to all users.
- Admins manage users, complaints, notifications, system analytics, and monitor transactions.

- **Out of Scope**

- Mobile app development (native iOS or Android apps)
- The system does not provide delivery management, tracking of delivery personnel, or take responsibility for item transportation.
- The system does not handle in-app payments, escrow services, or refunds, and all financial transactions are conducted outside the platform.
- The system only allows users to report disputes and does not provide any mediation or resolution mechanisms.
- The system does not provide insurance, enforce legal penalties, and functions solely as a facilitator between users.

Differences and Processes (Rent, Buying, sharing, donation)

Buying (Reselling)

The user lists an item for sale, another user searches and selects the item, sends a request, and once the seller accepts, both communicate and complete the transaction externally, after which the buyer receives the item and ownership is permanently transferred.

Buying results in permanent ownership transfer with payment.

Renting

The user lists an item for rent, another user searches and sends a request, and after the owner accepts, both agree on a rental period, the item is handed over, used temporarily, and then returned to the owner on the agreed date with condition verification.

Renting allows temporary use with payment, and the item must be returned.

Sharing (Free Lending)

The user lists an item for sharing, another user requests it, and after acceptance, the item is given for temporary use and later returned to the owner after the agreed time period.

Sharing allows temporary use without payment, and the item must be returned.

Donation

The user lists an item for donation, another user requests the item, and once the donor accepts, the item is handed over to the receiver, who keeps it permanently without any payment.

Donation results in permanent ownership transfer without payment.

4. Objectives of the project

Project Objectives

- Develop a functional web-based platform that enables users to exchange underutilized items through selling, renting, sharing, and donating.
- Implement core system features such as user management, item listing, search, request handling, and notifications.
- Ensure the system is user-friendly, reliable, and tested within the project timeline.
- Provide basic trust mechanisms such as ratings, reviews, and dispute reporting.
- Demonstrate the system's ability to facilitate successful item exchanges during testing.

Business Objectives

- Promote efficient utilization of underused resources by making them accessible to others.
- Reduce unnecessary purchasing by encouraging reuse, renting, and sharing.
- Provide a convenient platform for users to gain value from unused items.
- Support affordable access to goods for users who cannot purchase new items.
- Encourage a culture of sharing and responsible consumption within the community.

5. Project Feasibility

Technical Feasibility

The project is built using accessible, stable, and well-supported technologies without relying on frameworks or external libraries. The chosen tech stack includes:

- Frontend: HTML, CSS, JavaScript
- Backend: PHP
- Database: MySQL
- Server: Apache (via XAMPP or similar)
- Development Environment: Personal computers of the team members

All tools used are open-source or freely available for academic purposes. The development team has prior exposure to core web technologies and is continuously enhancing its skills in PHP and MySQL integration. The use of PHP ensures compatibility with standard web hosting environments and simplifies deployment. Given the team's capability and the appropriateness of the tools, the project is technically feasible.

Economic Feasibility

As a university-level academic project, development is carried out voluntarily by undergraduate students without financial compensation. Cost considerations include:

- Hosting: Development and testing on localhost via Apache server (XAMPP)
- Tools: Free and open-source software such as VS Code, MySQL, and PHP
- Hardware: Personal laptops and existing internet connections
- Domain: No domain purchase required during the development phase
- Personnel: Volunteer work by students as part of their coursework

Since no paid infrastructure or third-party services are involved, the financial cost is minimal, making the project economically feasible.

Social Feasibility

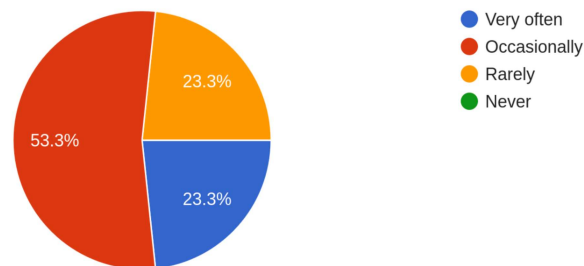
We conducted a survey using Google Forms to gather feedback from the general public across various regions in Sri Lanka. The survey received around 30 responses within a short period. Based on that we could see the approximate social feasibility.

Link:

<https://docs.google.com/forms/d/e/1FAIpQLSfTQNvJgVqJ9za5RUIQAYqEukR82AJDofWpiaM-ZNUKuV3c2g/viewform?usp=header>

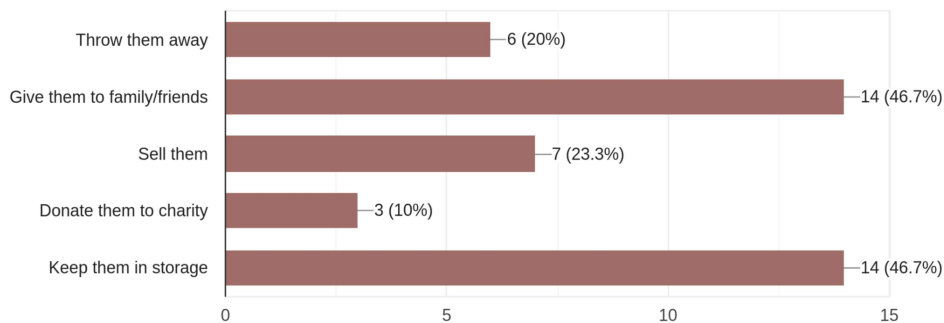
How often do you find yourself holding onto items you no longer use but are still in good condition?

30 responses



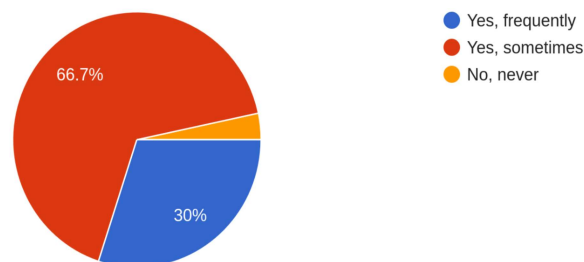
What do you currently do with items you no longer need?

30 responses



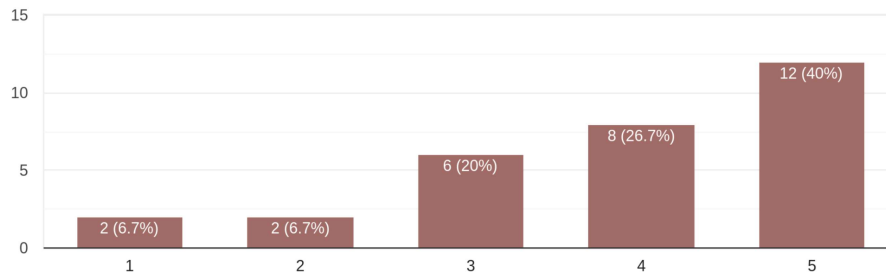
Have you ever wanted to use an item for a short period (e.g., camping gear, power tools, party supplies) but didn't want to buy it at full price?

30 responses



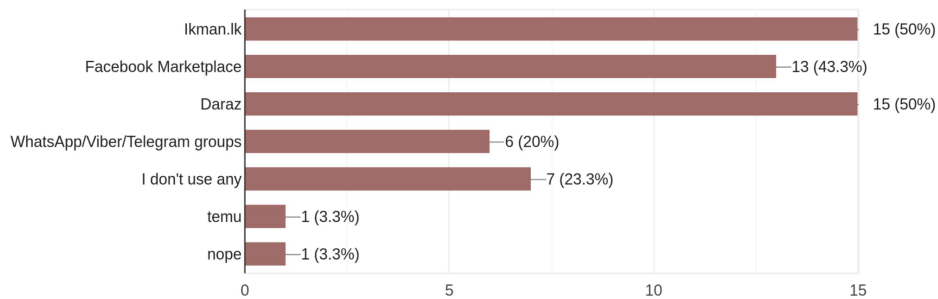
How likely are you to use a single web platform that allows you to seamlessly switch between selling, renting, and donating items?

30 responses



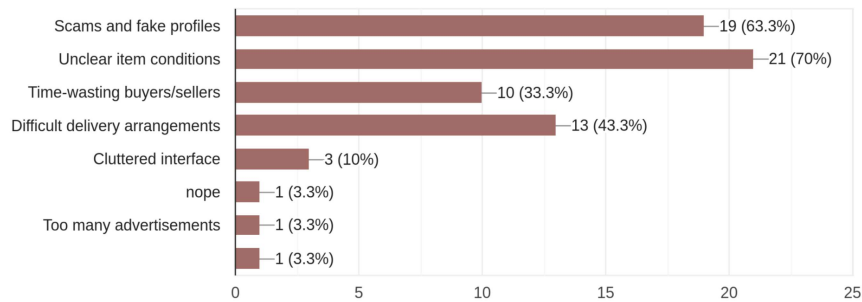
Which platforms do you currently use to buy, sell, or rent used items in Sri Lanka?

30 responses

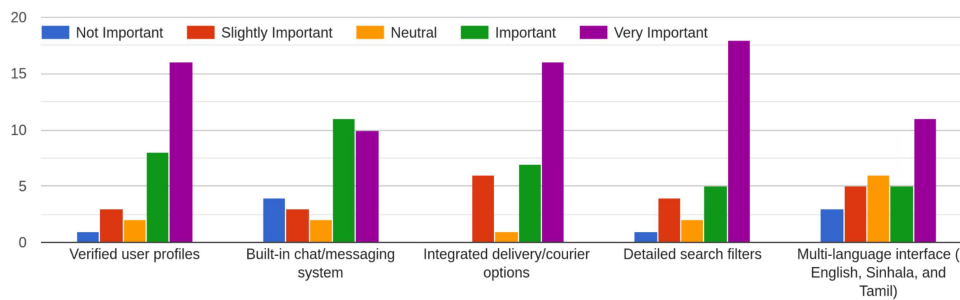


What frustrates you the most about the platforms you currently use?

30 responses

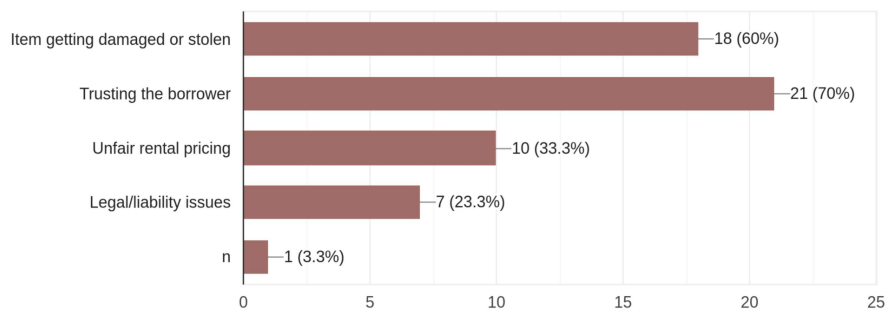


When using a platform like NexUse, how important are the following features to you?



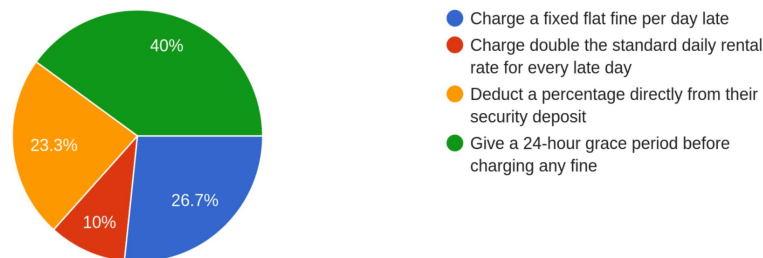
If you were to rent out your own personal items to someone else, what are your biggest concerns?

30 responses



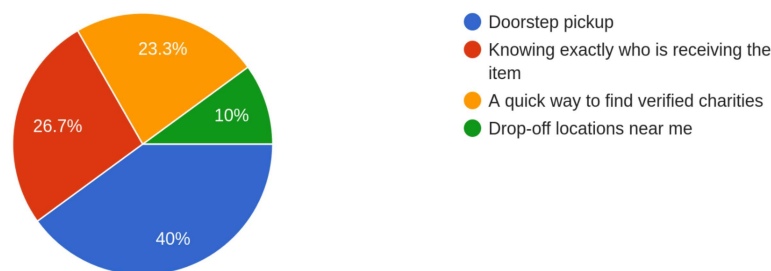
If a borrower returns a rented item late, what do you think is the fairest way to handle late penalties?

30 responses



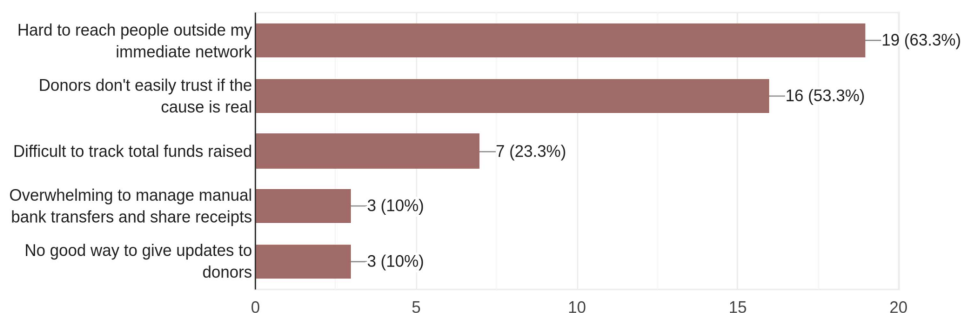
If you were to donate an item through NexUse, what would make the process most convenient for you?

30 responses



What is your biggest frustration when trying to raise donations using current methods (like posting on Facebook, WhatsApp status, or Instagram)?

30 responses



Legal and Ethical Feasibility

The initial version of the system does not handle sensitive financial or legal data. Nevertheless, basic user information is collected and managed responsibly.

Legal Considerations:

- Users are verified through admin approval
- User data is stored securely and accessed only by authorized roles
- No financial transactions or regulatory components are handled in this phase

Ethical Considerations:

- The platform supports price transparency and fair trading
- User data privacy is respected through basic security measures
- The system is designed to be inclusive and accessible for rural users

6. Deliverables of the project

Core Functional Deliverables (User Modules)

Module	Description
User Management Module	Registration, login/logout, profile management for all users.
Product Module	Sellers can add, edit, delete products with images, descriptions
Order Management Module	Buyers can browse, filter, and buy products.
Notifications Module	Trigger notifications (delivery status, responses) for Sellers, Donors, and Buyers
Complaints and Dispute Module	Buyers can submit complaints. Admin manages and resolves disputes between users.

Technical Deliverables

Backend System:

Developed using PHP.

Frontend:

Designed with HTML, CSS, and JavaScript (Vanilla) for responsive and user-friendly Interfaces.

Database Design and Implementation:

Structured using MySQL or PostgreSQL, with normalization and ER diagrams.

System Design Documentation:

Includes Use Case Diagrams, ER Diagrams, Sequence Diagrams, and Class Diagrams.

Testing Reports:

Covers unit testing, integration testing, and user acceptance testing (UAT).

Source Code Repository:

All source code maintained in GitHub, with clear version control and commit history.

UI Mockups:

Designed using Figma to visualize user interfaces and flows.

Deployment:

Deployed on a local server or free-tier cloud platform for demonstration.

Final Report and Presentation Materials:

Including the full project report, user manuals, and presentation slides for university evaluation.

7. Project Constraints and Assumptions

- The initial phase is restricted to a responsive web application.
- The system does not handle in-app payments.
- The system does not provide delivery management.

8. Requirements

Functional Requirements

Admin

- Log in and log out.
- View users by role.
- Edit or delete any user account.
- View a summary dashboard (total users, transactions, deliveries).
- View all product listings and transaction records.
- Manage and resolve disputes or complaints.
- Send notifications/messages to individual users or user groups.
- Generate reports (monthly, custom – e.g., sales, user activity).

Buyer

- Create a new buyer account by providing the required registration details.
- Access the system securely using a registered username/email and password.
- Maintain and update personal profile information whenever necessary.
- Browse available products listed by sellers.
- Search products using keywords.
- Filter products by category, price, location, or condition.
- View detailed product information including images and descriptions.
- Add products to a wishlist.
- Add products to the shopping cart.
- Place orders for selected products.
- View order status and purchase history.
- Communicate with sellers through the chat system.
- Rate and review products and sellers after successful transactions.

Seller

- Register an account to access seller features.
- Log in using valid credentials and log out.
- Manage and update profile information.
- Create product listings for items available for sale.
- Add and edit product descriptions, prices, and other relevant details.
- Delete or remove product listings.
- Receive notifications when new orders or inquiries are received.
- Communicate with buyers through the chat system.
- Report suspicious activities, fraudulent buyers, or scams.
- Verify account information to improve security and trustworthiness.

Donar

- Log in using valid credentials and log out.
- Manage and update their profile information.
- Browse active donation requests.
- Filter donation requests by category, location, or item type.
- If needed ,able to cancel a donation pledge before it is accepted.
- Can be able to view the status of pledged donations.
- Communicate with receivers.
- Can be able to view notifications related to donation activities.
- Can be able to view their donation history.
- Notify when a donation pledge is accepted or rejected.

Donation Receiver

- Log in using valid credentials and log out.
- Manage and update their profile information.
- Create and manage a donation request.
- Delete a donation request.
- Can be able to view donation pledges received for a request.
- Accept a donation pledge.
- Reject a donation pledge.
- Communicate with donors.
- Can be able to view notifications related to donation requests and pledges.
- Notify a requester when a donor pledges a donation.

Lender

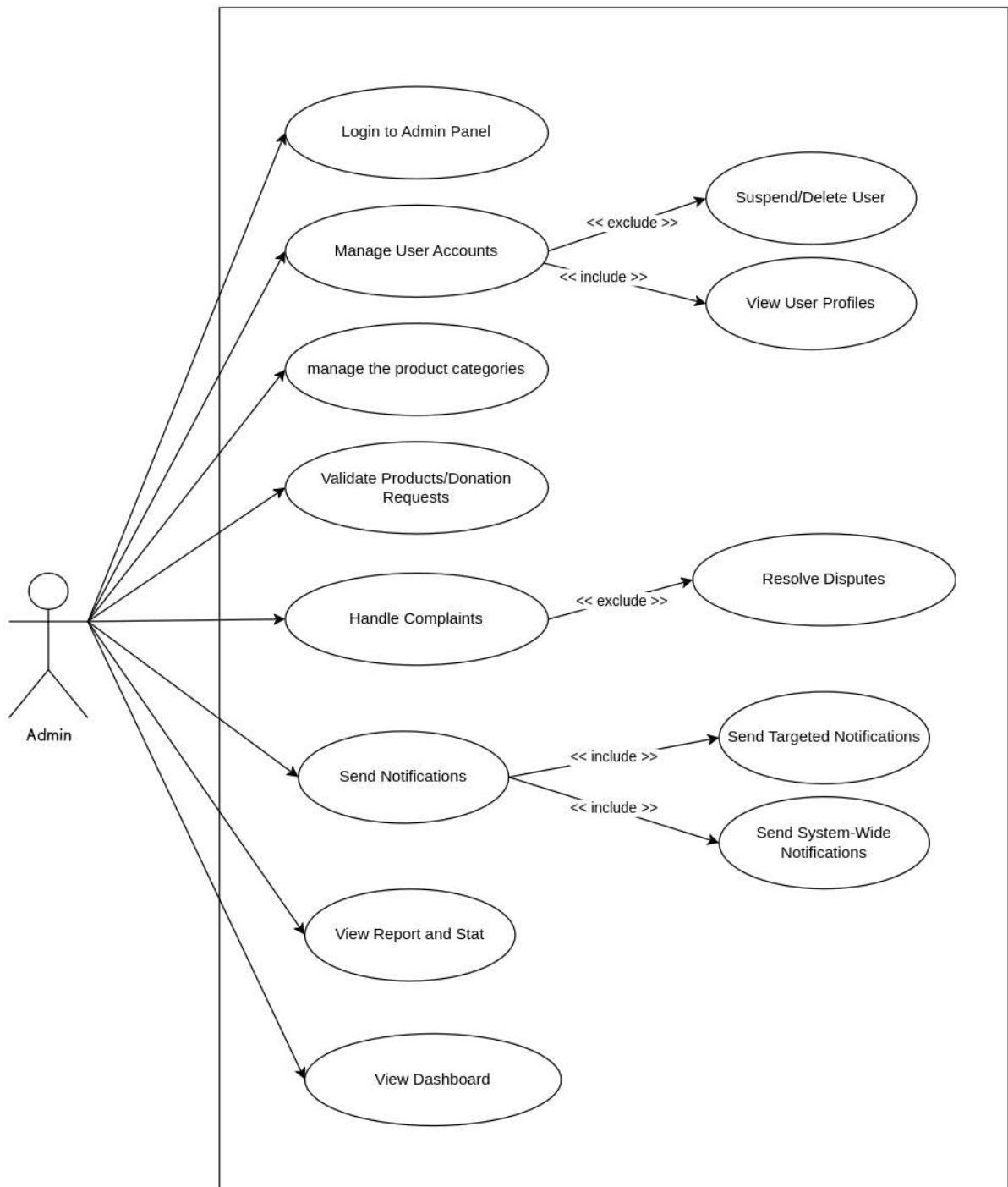
- Log in and log out.
- List items for rent with details (description, condition, price, availability).
- View and manage listed items.
- Receive rental requests from users.
- Accept or reject rental requests.
- Set rental duration (start date and return date).
- Track rented items and their status.
- Verify item condition upon return.
- Request compensation for damaged or unreturned items.
- View ratings and reviews from renters.

Render

- Log in and log out.
- Search and view items available for rent.
- Send rental requests for selected items.
- View status of rental requests (pending, accepted, rejected).
- View rental details (duration, conditions, return date).
- Receive notifications for request updates and return reminders.
- Use the item for the agreed rental period.
- Return the item within the agreed time.
- Confirm return of the item.
- Provide ratings and reviews for lenders.

Diagrams

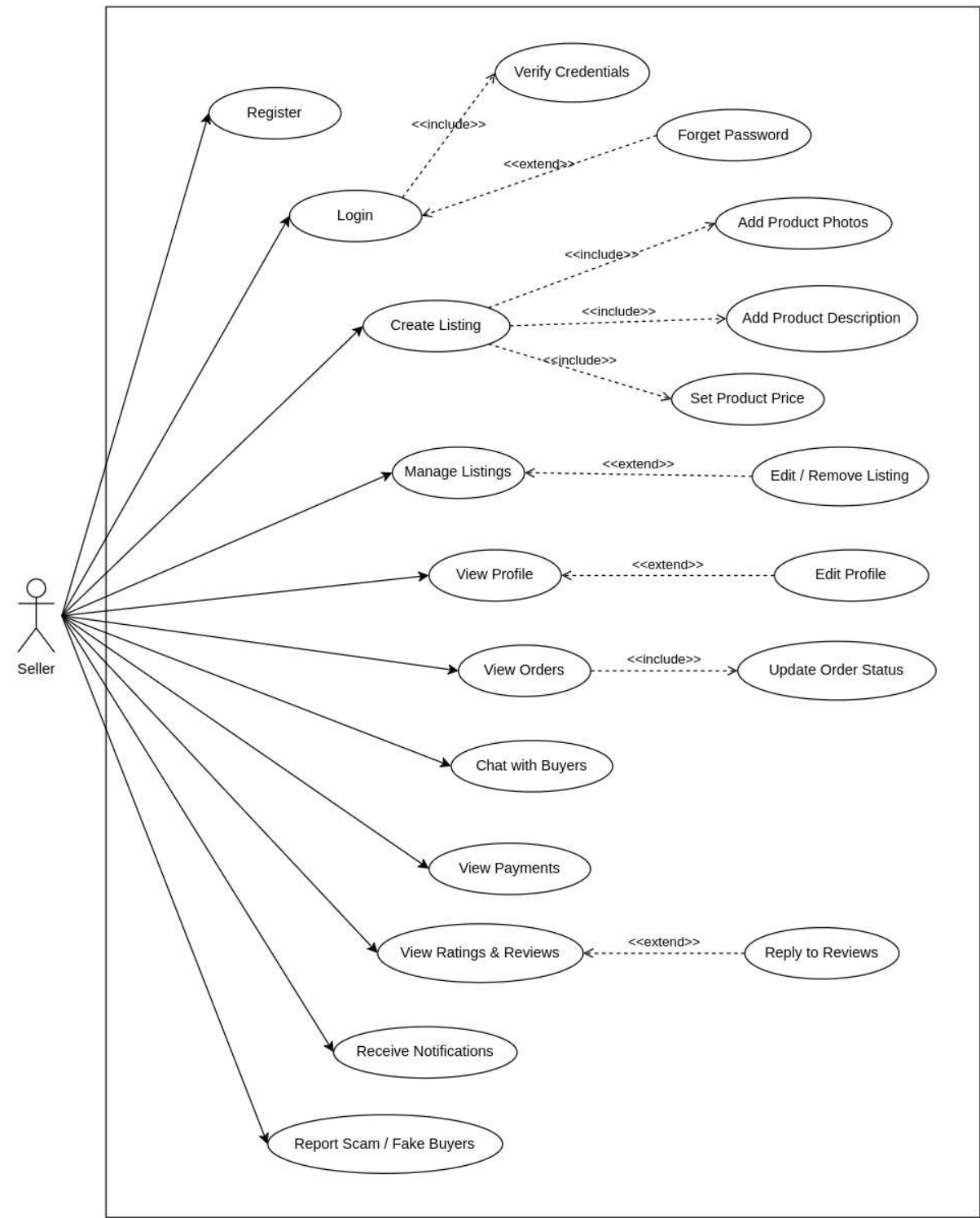
Admin



Buyer



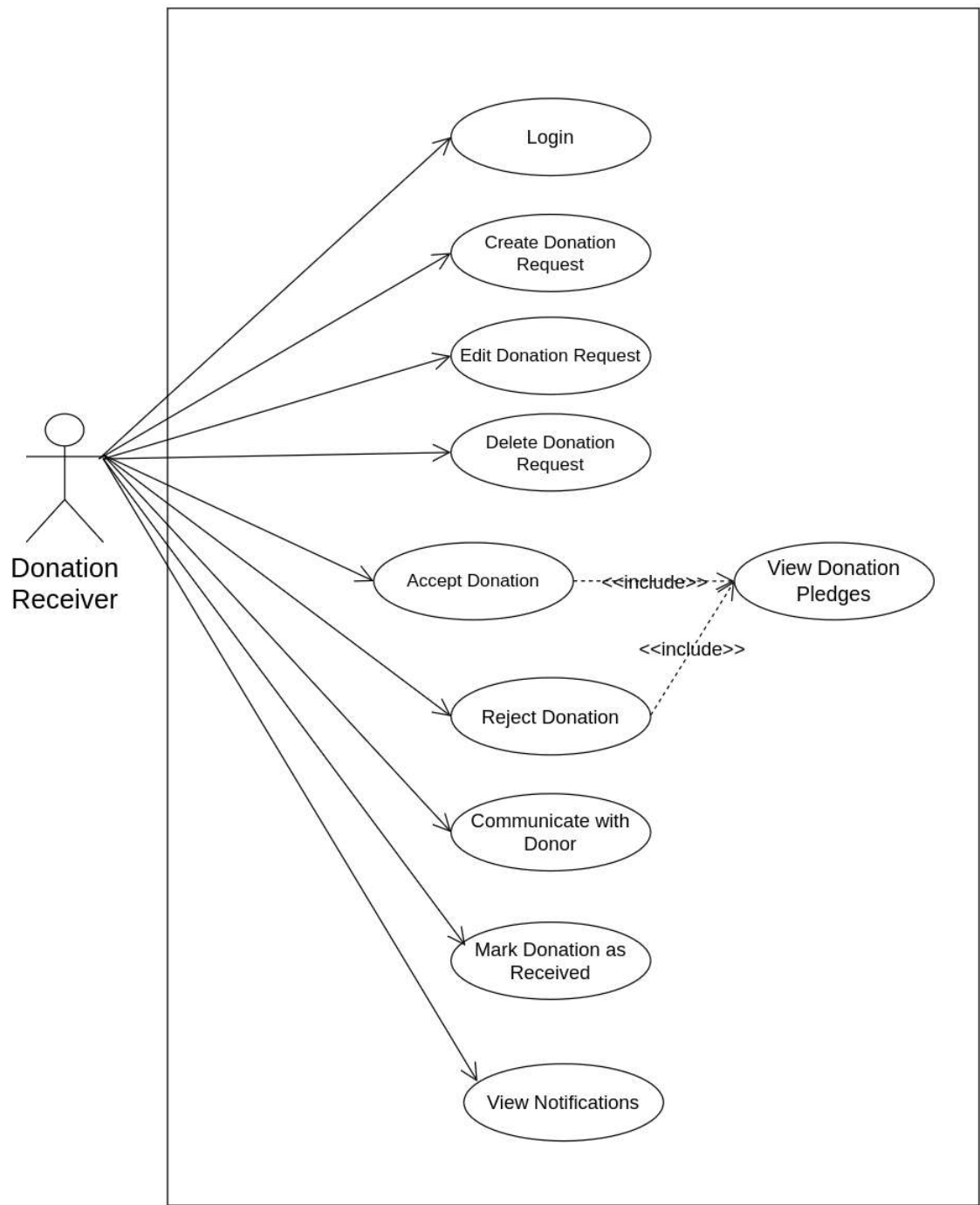
Seller



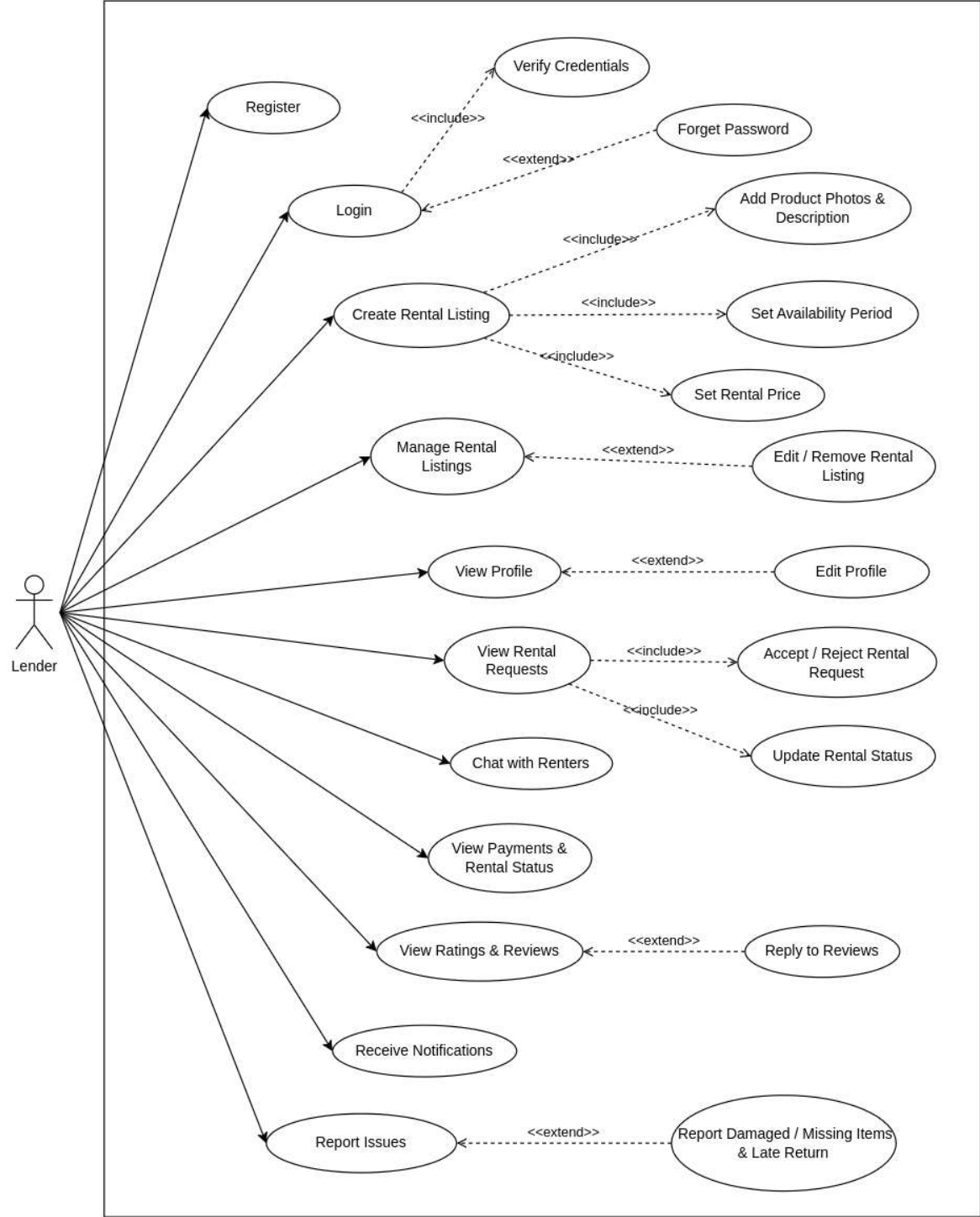
Donor



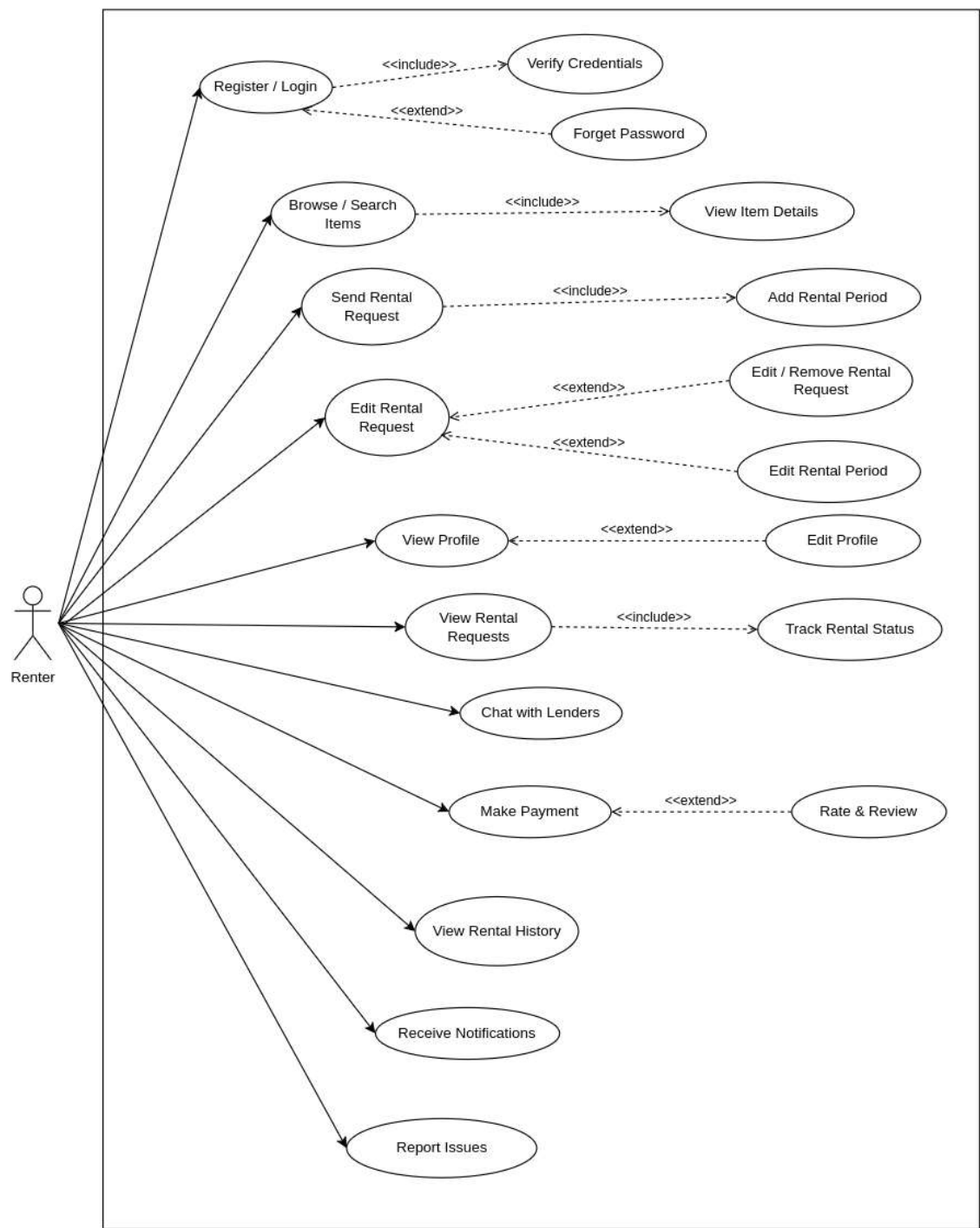
Donation Receiver



Lender



Renter



9. Technologies to be used

Our project will leverage a carefully selected suite of technologies tailored to ensure robust functionality, security, and scalability without reliance on additional frameworks.

Front End: HTML, CSS, JavaScript

- **HTML (HyperText Markup Language):**
Provides the structural foundation of the application, ensuring content is well-organized and accessible across devices. It is universally supported by all web browsers, ensuring that our legal case management system is accessible to users regardless of their preferred browser.
- **CSS (Cascading Style Sheets):**
Used for styling the user interface, ensuring a visually appealing and responsive design. It provides powerful tools for creating responsive designs that adapt to different screen sizes and devices.
- **JavaScript:**
Enables dynamic and interactive elements within the application. With JavaScript, we can implement features that improve the usability of our system. This includes creating interactive forms and providing instant search results. These enhancements make the system more intuitive and easier to navigate for users.

Back End: PHP

- **PHP:**
PHP integrates seamlessly with various database management systems, most notably MySQL, which is commonly used in web applications. This integration facilitates efficient data storage, retrieval, and management within our system, ensuring robust database interactions for storing crop listings, user information, and system configurations.

Database: MySQL

- **MySQL:**
A reliable and scalable relational database management system (RDBMS) ideal for handling structured data. It provides ACID compliance to ensure data integrity. It also supports complex queries and transactions necessary for managing large volumes of listings and information securely.

Designing tools: Figma and Canva

- **Figma:**
Figma enhances our ability to deliver a user-friendly and visually appealing interface for our system while maintaining efficiency and collaboration among our design team.
- **Canva:**
Canva enhances our ability to create engaging and professional visuals for our system, ensuring consistency and quality across our design assets

Development Tools and Deployment

- **Version Control:**
Git for managing codebase changes, facilitating collaboration among developers and maintaining version history
- **IDE (Integrated Development Environment):**
Visual Studio Code for code editing, enhancing productivity and code quality.